Data Structures and Algorithms

TP04: Linked List – Part 1

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01

tp4_ex1.cpp

Week04 > lab04 > tp4_ex1.cpp > ...

You, yesterday | 1 author (You)

```cpp
#include <iostream>
using namespace std;

// Node/Element structure for storing a single integer
struct Node {
    int data;        // Store the integer value
    Node* next;      // Pointer to the next node in the list
};

// List structure to manage the linked list
struct List {
    Node* head;      // Pointer to the first node
    Node* tail;      // Pointer to the last node
};

int main() {
    cout << "Exercise 1: Linked List Structure Definition" << endl;
    cout << "==========================================" << endl;
    cout << "\nStructures defined:" << endl;
    cout << "- Node: Contains data (int) and next pointer" << endl;
    cout << "- List: Contains head and tail pointers" << endl;
    cout << "\nThese structures can hold unlimited integer data." << endl;

    return 0;
}
```

02

tp4_ex2.cpp

Week04 > lab04 > tp4_ex2.cpp > ...

You, yesterday | 1 author (You)

```cpp
#include <iostream>
#include <string>
using namespace std;

// Student element structure
struct Student {
    int id;
    string name;
    int year;
    string programDegree;
    Student* next;           // Pointer to the next student node
};

// List structure for managing student data
struct ListStudent {
    Student* head;           // Pointer to the first student
    Student* tail;           // Pointer to the last student
};

int main() {
    cout << "Exercise 2: Student Linked List Structure Definition" << endl;
    cout << "==================================================" << endl;
    cout << "\nStructures defined:" << endl;
    cout << "- Student: Contains ID, name, year, programDegree, and next pointer" << endl;
    cout << "- ListStudent: Contains head and tail pointers" << endl;
    cout << "\nThese structures can store unlimited student data." << endl;

    return 0;
}
```

03

tp4_ex3.cpp

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You, yesterday | 1 author (You)

```cpp
#include <iostream>
#include <string>
using namespace std;

// Student element structure
struct Student {
    int id;
    string name;
    int year;
    string programDegree;
    Student* next;
};

// List structure for managing student data
struct ListStudent {
    Student* head;
    Student* tail;
};

// Function to create an empty list
ListStudent* createEmptyList() {
    ListStudent* list = new ListStudent();
    list->head = NULL;
    list->tail = NULL;
    return list;
}

// Function to add a student to the list
void add(Student s, ListStudent* list) {
    Student* newNode = new Student();
    newNode->id = s.id;
    newNode->name = s.name;
    newNode->year = s.year;
    newNode->programDegree = s.programDegree;
    newNode->next = NULL;

    if (list->head == NULL) {
        list->head = newNode;
        list->tail = newNode;
    } else {
        list->tail->next = newNode;
        list->tail = newNode;
    }
}

int main() {
```

```cpp
void add(Student s, ListStudent* list) {

int main() {
    cout << "Exercise 3: Create and Add Students to Linked List" << endl;
    cout << "==================================================" << endl << endl;

    // Create an empty list
    ListStudent* studentList = createEmptyList();

    // Create 5 student objects
    Student s1 = {1001, "Alice Johnson", 2, "Computer Science"};
    Student s2 = {1002, "Bob Smith", 2, "Software Engineering"};
    Student s3 = {1003, "Charlie Brown", 2, "Information Technology"};
    Student s4 = {1004, "Diana Prince", 2, "Data Science"};
    Student s5 = {1005, "Eve Wilson", 2, "Cybersecurity"};

    // Add all students to the list
    add(s1, studentList);
    add(s2, studentList);
    add(s3, studentList);
    add(s4, studentList);
    add(s5, studentList);

    cout << "5 students have been added to the linked list." << endl;
    cout << "Students: " << s1.name << ", " << s2.name << ", " << s3.name
         << ", " << s4.name << ", " << s5.name << endl;

    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS cppdbg

```
PS N:\School\Y2-S2\data-structures-and-algorithm>  & 'c:\Users\eangm\.vscode\extensions\ms-vscode.cpptools-1.31.5-win32-x64\debugAdapters\bin\
--stdin=Microsoft-MIEngine-In-kn3b5bph.mmf' '--stdout=Microsoft-MIEngine-Out-lvntzbte.cyc' '--stderr=Microsoft-MIEngine-Error-qdhxk1uq.ez4' '-
-dtg1tkx3.q3d' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Exercise 3: Create and Add Students to Linked List
==================================================

5 students have been added to the linked list.
Students: Alice Johnson, Bob Smith, Charlie Brown, Diana Prince, Eve Wilson
PS N:\School\Y2-S2\data-structures-and-algorithm>
```

04

tp4_ex4.cpp

Week04 > lab04 > tp4_ex4.cpp > ...

You, yesterday | 1 author (You)

```cpp
#include <iostream>
#include <string>
#include <iomanip>
using namespace std;

// Student element structure
struct Student {
    int id;
    string name;
    int year;
    string programDegree;
    Student* next;
};

// List structure for managing student data
struct ListStudent {
    Student* head;
    Student* tail;
};

// Function to create an empty list
ListStudent* createEmptyList() {
    ListStudent* list = new ListStudent();
    list->head = NULL;
    list->tail = NULL;
    return list;
}

// Function to add a student to the list
void add(Student s, ListStudent* list) {
    Student* newNode = new Student();
    newNode->id = s.id;
    newNode->name = s.name;
    newNode->year = s.year;
    newNode->programDegree = s.programDegree;
    newNode->next = NULL;

    if (list->head == NULL) {
        list->head = newNode;
        list->tail = newNode;
    } else {
        list->tail->next = newNode;
        list->tail = newNode;
    }
}
```

tp4_ex4.cpp

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```cpp
// Function to display all students in the list
void display(ListStudent* list) {
    cout << "\n========== STUDENT LIST ==========" << endl;
    cout << setw(8) << "ID" << setw(20) << "Name" << setw(6) << "Year"
         << setw(20) << "Program Degree" << endl;
    cout << "==================================" << endl;

    Student* current = list->head;
    int count = 0;

    while (current != NULL) {
        cout << setw(8) << current->id << setw(20) << current->name
             << setw(6) << current->year << setw(20) << current->programDegree << endl;
        current = current->next;
        count++;
    }

    cout << "==================================" << endl;
    cout << "Total students: " << count << endl << endl;
}

int main() {
    cout << "Exercise 4: Display Students and Add 2 More" << endl;
    cout << "==========================================" << endl;

    // Create an empty list
    ListStudent* studentList = createEmptyList();
```

tp4_ex4.cpp

Week04 > lab04 > tp4_ex4.cpp > ...

```cpp
int main() {
    // Create an empty list
    ListStudent* studentList = createEmptyList();

    // Add 5 initial students
    Student s1 = {1001, "Alice Johnson", 2, "Computer Science"};
    Student s2 = {1002, "Bob Smith", 2, "Software Engineering"};
    Student s3 = {1003, "Charlie Brown", 2, "Information Technology"};
    Student s4 = {1004, "Diana Prince", 2, "Data Science"};
    Student s5 = {1005, "Eve Wilson", 2, "Cybersecurity"};

    add(s1, studentList);
    add(s2, studentList);
    add(s3, studentList);
    add(s4, studentList);
    add(s5, studentList);

    // Display the list
    cout << "\nFirst, displaying the initial 5 students:" << endl;
    display(studentList);

    // Add 2 more students
    Student s6 = {1006, "Frank Miller", 2, "Artificial Intelligence"};
    Student s7 = {1007, "Grace Lee", 2, "Cloud Computing"};

    add(s6, studentList);
    add(s7, studentList);

    // Display all students including the 2 new ones
    cout << "After adding 2 more students:" << endl;
    display(studentList);

    return 0;
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS cppdbg:

```
PS N:\School\Y2-S2\data-structures-and-algorithm>  & 'c:\Users\eangm\.vscode\extensions\ms-vscode.cpptools-1.31.5-win32-x64\debugAdapters\bin\
--stdin=Microsoft-MIEngine-In-qb1fimib.xss' '--stdout=Microsoft-MIEngine-Out-nzjiyjs3.p3h' '--stderr=Microsoft-MIEngine-Error-ikh3bmaj.ker' '-
-dgf1kbin.kkp' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
----------------------------------
    1001       Alice Johnson     2    Computer Science
    1002           Bob Smith     2Software Engineering
    1003       Charlie Brown     2Information Technology
    1004        Diana Prince     2        Data Science
    1005          Eve Wilson     2       Cybersecurity
    1006        Frank Miller     2Artificial Intelligence
    1007           Grace Lee     2     Cloud Computing
==================================
Total students: 7

PS N:\School\Y2-S2\data-structures-and-algorithm>
```

Github Link:

https://github.com/mengly-eg/data-structures-and-algorithm.git